Bayesian Optimization Report: Function 4

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.617	Medium
2	0.619	Medium
3	0.622	Medium
4	0.624	Medium
5	0.625	Medium
6	0.627	Medium
7	0.629	Medium
8	0.631	Medium
9	0.633	Medium
10	0.634	Medium

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	1.6279	0.4185	3.8521
2	1.5812	0.5975	3.9655
3	1.5501	0.7381	3.4716
4	1.4630	0.5624	4.2259
5	1.4364	0.7012	3.6665
6	1.4604	0.5727	4.4586
7	1.4298	0.4013	3.6810
8	1.3678	0.5109	3.2879
9	1.3280	0.5372	3.5360
10	1.3612	0.5487	3.3904

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[1. 0.80765099 0.86224114 1.]	-43.6443233996 3481	Matern(ν=0.5)	EI	0.617	None

2	[1. 0.82755924 0.98053371 0.9500518]	-45.8824207634779	Matern(ν=0.5)	EI	0.619	None
3	[0.97733676 0.9910808 0.66762897 0.97332556]	-42.60753920861701	Matern(ν=0.5)	EI	0.622	None
4	[1. 0.42891627 0.87219738 1.]	-34.80751975548477	Matern(ν=0.5)	EI	0.624	None
5	[1. 0.74735887 1. 1.]	-45.392629858838845	Matern(ν=0.5)	EI	0.625	None
6	[1. 0.92120991 0.92324764 1.]	-48.33382592385269	Matern(ν=0.5)	EI	0.627	None
7	[1. 0.99211713 1. 0.84769781]	-47.917066118891555	Matern(ν=0.5)	EI	0.629	None
8	[1. 0.83003791 0.87373381 0.94076116]	-42.87536530757693	Matern(ν=0.5)	EI	0.631	None
9	[0.55544196 1. 0.89271657 1.]	-40.2071350194812	Matern(ν=0.5)	EI	0.633	None
10	[1. 0.46361266 0.88636525 0.9714829]	-34.905288332667354	Matern(ν=0.5)	EI	0.634	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	1.110008
EI	0.019397
PI	0.061760
THOMPSON	0.322230
KG	0.019397
ENTROPY	inf
PORTFOLIO	2.095000

Iteration 2

Method	Max Value
UCB	-0.066710
EI	0.003306
PI	0.011463
THOMPSON	-0.266326
KG	0.003306
ENTROPY	inf

PORTFOLIO	1.552006

Iteration 3

Method	Max Value
UCB	-0.170034
EI	0.001344
PI	0.005448
THOMPSON	-0.042951
KG	0.001344
ENTROPY	406.615132
PORTFOLIO	1.142826

Iteration 4

Method	Max Value
UCB	0.213638
EI	0.008082
PI	0.032487
THOMPSON	-0.224225
KG	0.008082
ENTROPY	inf
PORTFOLIO	1.127340

Iteration 5

Method	Max Value
UCB	0.045502
EI	0.004849
PI	0.017343
THOMPSON	-0.183542
KG	0.004849
ENTROPY	inf
PORTFOLIO	1.145289

Iteration 6

Method	Max Value
UCB	-0.122456

EI	0.001955
PI	0.007555
THOMPSON	-0.870613
KG	0.001955
ENTROPY	inf
PORTFOLIO	1.010086

Iteration 7

Method	Max Value
UCB	0.192523
EI	0.002678
PI	0.017422
THOMPSON	-0.122258
KG	0.002678
ENTROPY	inf
PORTFOLIO	0.769865

Iteration 8

Method	Max Value
UCB	-0.125087
EI	0.002133
PI	0.008011
THOMPSON	-0.737912
KG	0.002133
ENTROPY	inf
PORTFOLIO	0.619965

Iteration 9

Method	Max Value
UCB	0.459293
EI	0.030855
PI	0.107522
THOMPSON	0.395691
KG	0.030855
ENTROPY	inf

PORTFOLIO	1.015905

Iteration 10

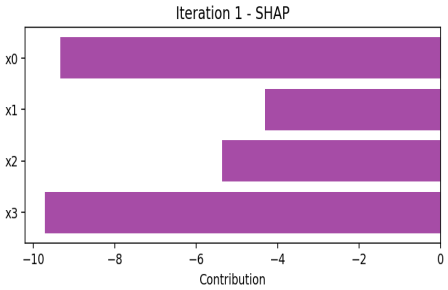
Method	Max Value
UCB	0.255370
EI	0.010553
PI	0.040672
THOMPSON	0.466794
KG	0.010553
ENTROPY	inf
PORTFOLIO	0.453912

LOO MSE Summary

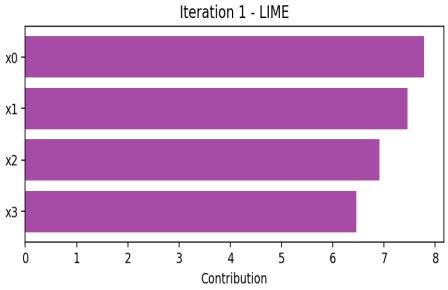
Iteration	LOO MSE
1	4.830085
2	3.861362
3	1.997051
4	1.894813
5	1.749928
6	1.730586
7	1.618786
8	2.887540
9	2.808089
10	2.525324

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



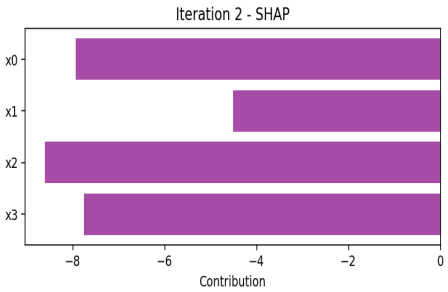
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	-9.34074
x1	-4.31236
x2	-5.36525
x3	-9.71309

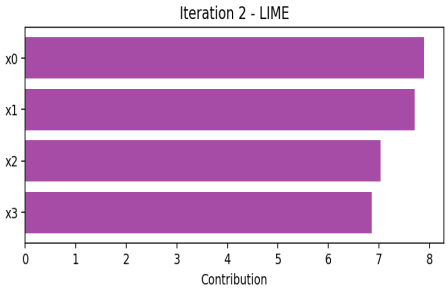
Iteration 1 - LIME Values

Feature	LIME Value
x0	7.77839
x1	7.45628
x2	6.90985
x3	6.46024

Iteration 2 - SHAP



Iteration 2 - LIME



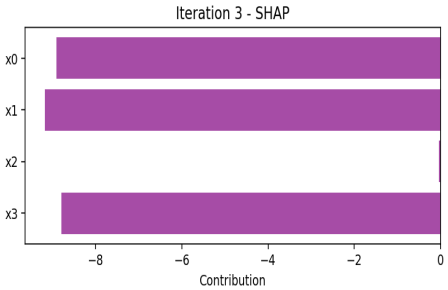
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	-7.93544
x1	-4.51186
x2	-8.60064
x3	-7.74949

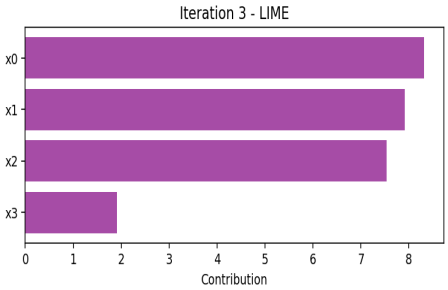
Iteration 2 - LIME Values

Feature	LIME Value
x0	7.89059
x1	7.70813
x2	7.03458
x3	6.85659

Iteration 3 - SHAP



Iteration 3 - LIME



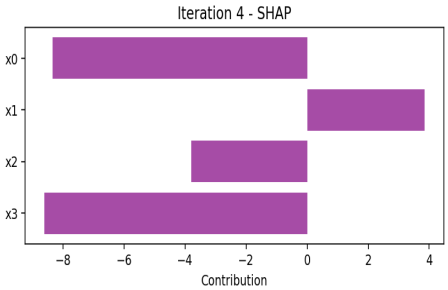
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	-8.90756
x1	-9.17408
x2	-0.02955
x3	-8.79063

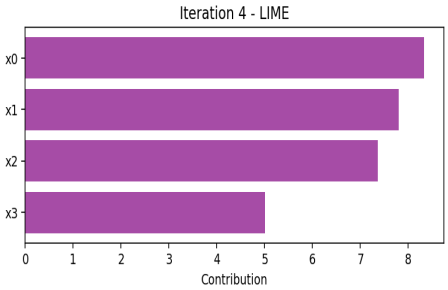
Iteration 3 - LIME Values

Feature	LIME Value
x0	8.31563
x1	7.92104
x2	7.53568
x3	1.91430

Iteration 4 - SHAP



Iteration 4 - LIME



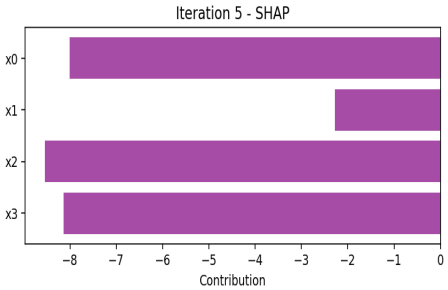
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-8.33812
x1	3.85412
x2	-3.79846
x3	-8.61223

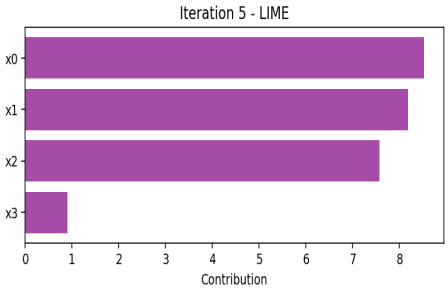
Iteration 4 - LIME Values

Feature	LIME Value
x0	8.32848
x1	7.80468
x2	7.36737
x3	5.00727

Iteration 5 - SHAP



Iteration 5 - LIME



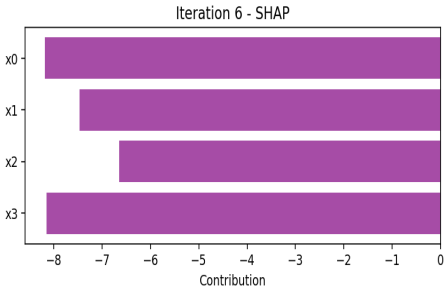
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	-8.00839
x1	-2.28150
x2	-8.53744
x3	-8.13396

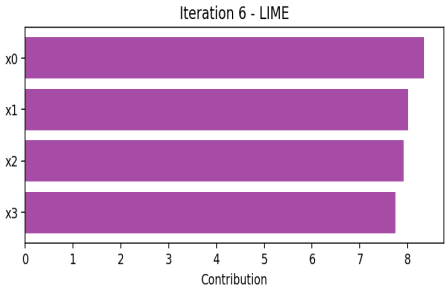
Iteration 5 - LIME Values

Feature	LIME Value
x0	8.52128
x1	8.18704
x2	7.57519
x3	0.90436

Iteration 6 - SHAP



Iteration 6 - LIME



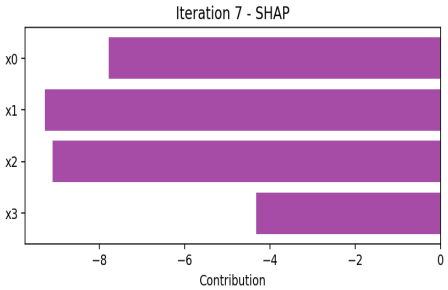
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	-8.18324
x1	-7.46800
x2	-6.64656
x3	-8.15836

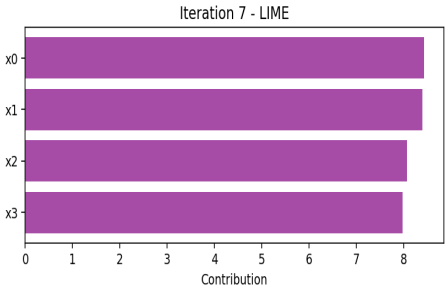
Iteration 6 - LIME Values

Feature	LIME Value
x0	8.34274
x1	8.01088
x2	7.91483
x3	7.74489

Iteration 7 - SHAP



Iteration 7 - LIME



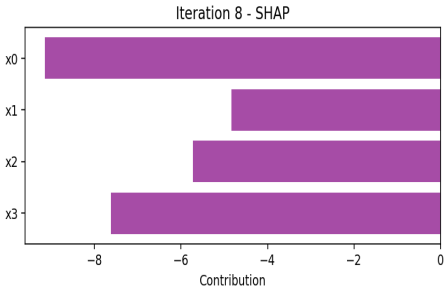
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	-7.78170
x1	-9.27288
x2	-9.09873
x3	-4.32482

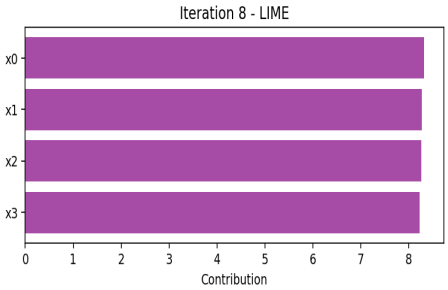
Iteration 7 - LIME Values

Feature	LIME Value
x0	8.43302
x1	8.40594
x2	8.07492
x3	7.98421

Iteration 8 - SHAP



Iteration 8 - LIME



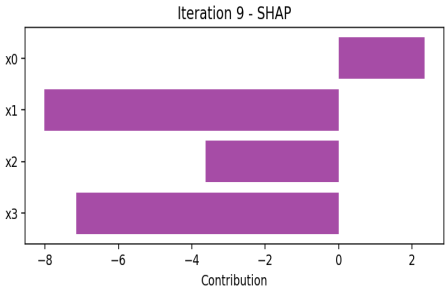
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	-9.14435
x1	-4.82840
x2	-5.72411
x3	-7.62233

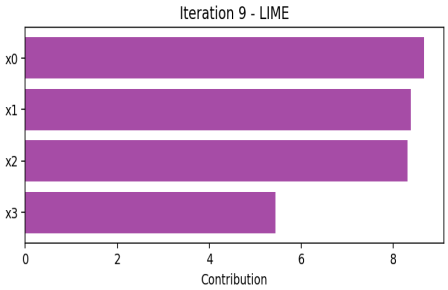
Iteration 8 - LIME Values

Feature	LIME Value
x0	8.31793
x1	8.28095
x2	8.26882
x3	8.22769

Iteration 9 - SHAP



Iteration 9 - LIME



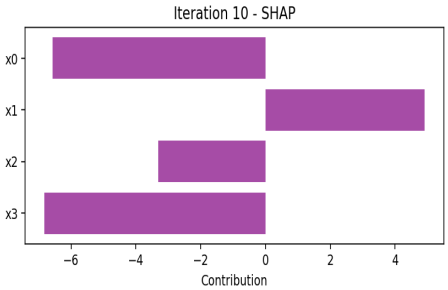
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	2.35127
x1	-8.02058
x2	-3.62539
x3	-7.15646

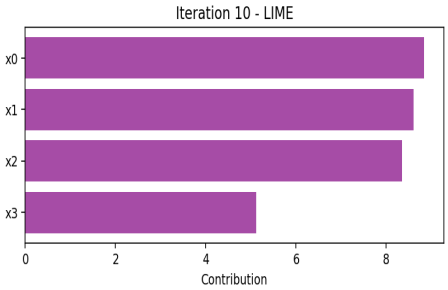
Iteration 9 - LIME Values

Feature	LIME Value
x0	8.66246
x1	8.37400
x2	8.30948
x3	5.44136

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	-6.56844
x1	4.91914
x2	-3.30509
x3	-6.82620

Iteration 10 - LIME Values

Feature	LIME Value
x0	8.83476
x1	8.60135
x2	8.35176
x3	5.11640